1 Vectors

A vector $\mathbf{v} \in \mathbb{R}^n$ is a list of n real numbers. Vectors of the same length are added component by component, and a vector is scaled by multiplying every component by the same number. The dot product of two vectors combines them into a single number:

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i. \tag{1.1}$$

The length of a vector is the square root of its dot product with itself, so $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$. Two vectors are orthogonal when their dot product is zero. In mechanics, a rigid body turning with angular velocity $\boldsymbol{\omega}$ moves a point at position $\mathbf{r}$ with velocity

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}. \tag{1.2}$$

2 Matrices

A matrix with m rows and n columns maps vectors of $\mathbb{R}^n$ to vectors of $\mathbb{R}^m$. For a two by two matrix the determinant is easy to compute:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad \det A = -2. \tag{2.1}$$

A linear system collects several equations into one matrix equation $A\mathbf{x} = \mathbf{b}$. For example,

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}. \tag{2.2}$$

When $\det A \neq 0$ the system has exactly one solution, $\mathbf{x} = A^{-1}\mathbf{b}$.

3 Eigenvalues

A number λ is an eigenvalue of A when $A\mathbf{x} = \lambda\mathbf{x}$ for some nonzero vector $\mathbf{x}$. To find the eigenvalues:

(i) form the matrix $A - \lambda I$;

(ii) solve $\det(A - \lambda I) = 0$ for λ;

(iii) for each root, solve $(A - \lambda I)\mathbf{x} = \mathbf{0}$.

For the matrix in Equation (2.1) the characteristic polynomial is

$$\lambda^2 - 5\lambda - 2 = 0, \tag{3.1}$$

whose roots are $\lambda = \frac{5 \pm \sqrt{33}}{2}$. The sum of the eigenvalues is the trace of the matrix and their product is its determinant, which gives a quick check of the arithmetic. Symmetric matrices have real eigenvalues and orthogonal eigenvectors, a fact used again and again in the rest of the course. They also make the next topic, projecting a vector onto a line, much easier to compute by hand.

4 Projections

The projection of a vector $\mathbf{b}$ onto the line through a nonzero vector $\mathbf{a}$ is the multiple of $\mathbf{a}$ closest to $\mathbf{b}$:

$$\operatorname{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a}. \tag{4.1}$$

The error $\mathbf{b} - \operatorname{proj}_{\mathbf{a}} \mathbf{b}$ is orthogonal to $\mathbf{a}$. Three facts follow:

- projecting twice gives the same vector as projecting once;
- the length of a projection is never more than the length of $\mathbf{b}$;
- the projection of $\mathbf{a}$ itself is $\mathbf{a}$.

Least squares fits use the same idea with a whole subspace in place of the line, and the normal equations $A^{\mathsf{T}}A\hat{\mathbf{x}} = A^{\mathsf{T}}\mathbf{b}$ give the best approximate solution of a system with no exact one.